CoEvolution

A Comprehensive Trustworthy Framework for Connected Machine Learning and Secure Interconnected AI Solutions

Work Package WP1

Project Management

Deliverable D1.2

Data management plan and a plan for legal and ethical requirements analysis (living document)

Project Acronym:	CoEvolution
Grant agreement No:	101168560
Funding Scheme:	HORIZON Research and Innovation Actions
Call:	HORIZON-CL3-2023-CS-01
Call Topic:	HORIZON-CL3-2023-CS-01-03 - Security of robust AI systems
Project Start:	1 November 2024
Project Duration:	36 months
Lead Participant:	AEGIS

This project has received funding from the European Union's Horizon Europe Framework Programme (HORIZON) under Grant Agreement No. 101168560.

Deliverable Information

Deliverable Administrative Information	
Project Acronym:	CoEvolution
Project Number:	101168560
Deliverable Number:	D1.2
Deliverable Full Title:	Data management plan and a plan for legal and ethical requirements analysis (living document)
Deliverable Short Title:	Data Management Plan
Document Identifier:	CoEvolution-D12-DataManagementPlan-submitted
Lead Participant:	AEGIS
Version:	v1.0
Due Date:	30/04/2025
Submission Date:	29/04/2025
Dissemination Level:	SEN
Deliverable Type:	Data Management Plan
Lead Editor(s):	AEGIS
Author(s):	AEGIS
Reviewers:	CBRL, AVL
Keywords:	Data Management Plan, Legal requirements, Ethical requirements
Status:	( ) draft, ( ) final, (✓) submitted

Deliverable Revisions

Date	Version	List of Contributor(s)	Summary of Changes
07/03/2025	v0.1	AEGIS	ToC drafted, Input solicited
28/03/2025	v0.2	AEGIS	Initial input for every Section
15/04/2025	v0.3	AEGIS	Finalised the content in each Section
23/04/2025	v0.4	CBRL, AVL	Draft, internally reviewed
28/04/2025	v0.5	AEGIS	Submitted the updated version to the reviewers, WPL, SPM, and PC, awaiting their approval
29/04/2025	v1.0	ICCS	Final editorial check and submit

Consortium

Part. No.	Participant Organization Name	Part. Short Name	Country
1	EREVNITIKO PANEPISTIMIAKO INSTITOUTO SYSTIMATON EPIKOINONION KAI YPOLOGISTON	ICCS	GREECE
2	ATHINA-EREVNITIKO KENTRO KAINOTOMIAS STIS TECHNOLOGIES TIS PLIROFORIAS, TON EPIKOINONION KAI TIS GNOSIS	ISI	GREECE
3	UNIVERSITA DEGLI STUDI DI CAGLIARI	UNICA	ITALY
4	SIEMENS AKTIENGESELLSCHAFT	SIE	GERMANY
5	AEGIS IT RESEARCH GMBH	AEGIS	GERMANY
6	TELEFONICA INNOVACION DIGITAL SL	TID	SPAIN
7	UNIVERSITA DI PISA	UNIPI	ITALY
8	SECURITY LABS CONSULTING LIMITED	SLC	IRELAND
9	AVL LIST GMBH	AVL	AUSTRIA
10	AVISENSE.AI TECHNOVLASTOS IKE	AVI	GREECE
11	DIINEKES S.I. MONOPROSOPI IDIOTIKI KEFALAIOUCHIKI ETAIREIA	DIE	GREECE
12	CYBERALYTICS LIMITED	CBRL	CYPRUS
13	SPHYNX TECHNOLOGY SOLUTIONS AG	STS	SWITZERLAND

Disclaimer for Deliverables

Table of Contents

List of Tables

List of Abbreviations

AAS	Asset Administration Shell
AI	Artificial Intelligence
ALTAI	Assessment List for Trustworthy Artificial Intelligence
API	Application Programming Interface
DMP	Data Management Plan
DOI	Digital Object Identifier
DPO	Data Protection Officer
EC	European Commission
FAIR	Findable Accessible Interoperable Reusable
GA	Grant Agreement
GDPR	General Data Protection Regulation
MFA	Multi-Factor Authentication
PC	Project Coordinator
PET	Privacy Enhancing Technology
TLS	Transport Layer Security
UUID	Universally Unique Identifier
WPs	Work Packages

Executive Summary

This deliverable introduces the Data Management Plan (DMP) for the CoEvolution project, prepared at M6 of the project following the European Commission's Guidelines for the Horizon Europe Programme. These guidelines mandate the submission of a DMP within the initial six months of the project. This report is the initial DMP for CoEvolution, reflecting the project's technical advancements at the time of writing. The DMP will be updated as the project evolves to include new developments and insights.

In adherence to the Findable Accessible Interoperable Reusable (FAIR) Principles, this deliverable outlines how research data collected and generated during the project will be made FAIR. It covers data management during and after the project, the standards and methods used for data collection and generation, and the data sharing approach. The framework for this document is based on the corresponding template provided by the European Commission (EC).

The submission of this document is part of WP1 activities, which continue until M36 of the CoEvolution project. As project activities develop, the DMP will remain a living document and be updated as needed to reflect ongoing progress.

1 Introduction

This document, the DMP for the CoEvolution project, is a comprehensive roadmap outlining the strategies and procedures for handling the data generated, collected, and used throughout the course of the project. It addresses key aspects such as data types, collection and processing methodologies, data sharing and preservation strategies, ethical and legal considerations, and the application of Artificial Intelligence (AI). The DMP underscores the project's commitment to rigorous data management, privacy, security, compliance with the prevailing legislations, and fostering a transparent and collaborative research environment. As a living document, it will be updated at M12 and M18 to capture planned developments and any new requirements identified through ongoing work. Additionally, interim updates will be made as necessary whenever significant changes occur in the project's activities, objectives, or technical developments. This approach will ensure that the DMP remains accurate, relevant, and aligned with best practices in data handling and compliance.

1.1 Purpose and Scope of the Document

The primary objective of this DMP is to outline a comprehensive framework for handling the research data generated, collected, and processed during the CoEvolution project. By doing so, the DMP aims to facilitate data sharing, dissemination, and long-term preservation, while promoting adherence to legislative, ethical, and privacy requirements. The DMP will guide project partners in managing data efficiently and responsibly throughout the project's lifecycle and beyond.

The scope of the DMP encompasses all aspects of data management within the CoEvolution project, including the following:

- Identifying the types and formats of data generated, collected, and processed within the project.
- Establishing procedures for data storage, backup, and security.
- Ensuring compliance with legal and ethical requirements, including General Data Protection Regulation (GDPR) and relevant legislation.
- Promoting adherence to the Findable, Accessible, Interoperable, and Reusable (FAIR) data principles.
- Defining data sharing and dissemination policies, as well as provisions for data access and licensing.
- Outlining the responsibilities of project partners in managing and maintaining data quality.
- Detailing the strategies for long-term data preservation and reusability.

The DMP will be reviewed and updated at regular intervals throughout the project, reflecting the evolving needs and progress of the CoEvolution project. This iterative approach ensures that the DMP remains a relevant and useful resource for all project partners, enabling effective data management and promoting research excellence.

1.2 Contribution to other Deliverables

The DMP is fundamentally linked with all the CoEvolution project's Work Packages (WPs) and their corresponding deliverables. This relationship results from the pervasive nature of data management, which pervades and influences all aspects of the project, from coordination and execution to dissemination, exploitation, and standardization. Consequently, the DMP serves not only as a central reference document but also as a strategic guide for these interconnected project components.

Below are the principal CoEvolution project elements that interact directly with the DMP:

- **All Work Packages:** Every work package, including tasks related to project coordination, dissemination, exploitation, and standardization, involves some degree of data usage. As such, the DMP's provisions form a foundational framework guiding the management of various data used throughout the project lifecycle. This interaction ensures consistent, quality, and compliant data handling practices across all project activities.
- **Technical Work Packages:** In addition to the overarching role, the DMP plays a specific role in technical work packages, which involve the generation, collection, and processing of data. The DMP outlines the standards and procedures for these activities, reinforcing data quality and security.
- **Ethical Guidelines:** The ethical guidelines defined for the CoEvolution project provide an overarching framework for all data management activities, with particular attention given to handling confidential data. The DMP ensures that these guidelines are upheld across all its provisions.
- **Dissemination and Communication Activities:** These activities rely on data generated and processed within the project for effective execution. The DMP informs these activities by outlining the protocols for data sharing and dissemination, ensuring adherence to the stipulated data policies.
- **Exploitation Activities:** The exploitation of project outcomes often involves the use of project data. As such, the DMP's guidelines on the accessibility, reusability, and interoperability of data directly facilitate the efficient exploitation of project results.
- **Standardization Activities:** The DMP establishes the standards for data management within the project. This role directly feeds into standardization activities, providing a consistent and uniform approach to data handling.

The DMP remains a dynamic and integral part of the CoEvolution project by reflecting updates corresponding to changes in the project's activities, progress, and technical developments. This approach ensures that effective and compliant data management is maintained throughout the project duration.

1.3 Structure of the Document

This document is designed to guide stakeholders through the project's Data Management Plan strategies. It is structured as follows:

- Section 1 sets the plan's context, goals, scope, and interplay with other project deliverables.

- Section 2 provides a comprehensive overview of datasets to be used, including their types, characteristics, and relevance to various project activities.
- Section 3 explores the implementation of FAIR principles, detailing strategies to ensure data is findable, accessible, interoperable, and reusable.
- Section 4 describes the resources allocated for data management within the project.
- Section 5 outlines the security measures to protect data integrity and prevent unauthorised access.
- Section 6 focuses on ethical considerations and legal compliance, especially GDPR, and the impact of AI on data management.
- Section 7 summarises the DMP's key points, highlighting its role in guiding the project.

Finally, the Questionnaire shared with the partners is presented in Appendix A CoEvolution Data Management Plan Questionnaire v1.0. Each section, with its detailed subsections, ensures a thorough understanding of the project's data management approach, fostering a coherent and efficient system throughout its lifecycle.

2 Data Summary

2.1 Overview of datasets

CoEvolution encompasses an approach that involves generating, utilising, and managing diverse datasets. At this stage (M6), we provide a high-level overview of these datasets, as presented in Table 1. However, it is imperative to recognise that the specifics concerning these datasets, including their nature, extent, and utilisation, are subject to continuous refinement as the project unfolds. Each dataset plays a pivotal role in advancing CoEvolution's overarching objectives.

Table 1: CoEvolution Datasets

No	Name of Dataset	Responsible Partner	Description of Accessibility
01	Dataset(s) for AI Model Evaluation, Training Enhancement, and Data Perturbation and Augmentation	ICCS, DIE	Open Access + Restricted Access
02	Dataset(s) for Continuous Learning, and training Robust and Trustworthy ML Models	UNICA, UNIPI	Open Access
03	Dataset(s) for Context Awareness	TID	Open Access + Restricted Access
04	Dataset(s) for Anonymising Algorithms	CBRL	Open Access
05	Dataset(s) for Vulnerabilities Identification	SLC	Restricted Access
06	Dataset(s) for Pilot 1	SIE, AVL	Open Access + Restricted Access
07	Dataset(s) for Pilot 2	AVI	Open Access
08	Administrative Data	ICCS	Restricted Access
09	Dissemination & Communication Data	AEGIS	Open Access + Restricted Access
10	Exploitation Data	AVL	Restricted Access
11	Standardisation Data	CBRL	Open Access

2.2 Datasets descriptions

It is important to note that the details of the datasets described in the following subsections—including their scope and purpose—are still under discussion and subject to change. The project partners, many of whom will provide data, have made an initial identification of the datasets likely to be generated or used during the project. However, these are based on the current understanding of the project scope and may change as the project develops and specific requirements become clearer.

Ongoing discussions on architecture, specifications, and the changing cybersecurity landscape highlight the evolving nature of these datasets. As such, the details in the following subsections are provisional. Any changes will be clearly documented in future updates of this deliverable.

2.2.1 Dataset(s) for AI Model Evaluation, Training Enhancement, and Data Perturbation and Augmentation

Table 2: Dataset(s) for AI Model Evaluation, Training Enhancement, and Data Perturbation and Augmentation

Name of Dataset	Dataset(s) for AI Model Evaluation, Training Enhancement, and Data Perturbation and Augmentation
Accessibility	Open Access + Restricted Access for some datasets due to contractual reasons
Purpose and relation to the objectives	To train AI models, create new synthetic datasets, test hypotheses, and support conclusions
Data types	Text, Images, Numeric Data
Data Formats	Plain Text, CSV, Image Formats
Reuse of existing data	Yes, leveraging publicly available datasets
Data origin	Publicly available dataset or repository, privately owned dataset or repository, previously generated by the project team
Expected size of the data	Less than 1 TB
Data utility	Supporting AI model development and evaluation by addressing key research questions related to potential attacks on AI models, such as those in the training (data poisoning) and inference phases (adversarial examples). It will also help identify threats during data collection, ensuring data integrity and security, and guide the application of preprocessing techniques to mitigate data leakage and protect sensitive information.
Privacy Principles	Data will be reviewed to ensure compliance with ethical guidelines (e.g., privacy and consent) and legal restrictions (e.g., licensing or intellectual property rights).

2.2.2 Dataset(s) for Continuous Learning, and training Robust and Trustworthy ML Models

Table 3: Dataset(s) for Continuous Learning, and training Robust and Trustworthy ML Models

Name of Dataset	Dataset(s) for Continuous Learning, and training Robust and Trustworthy ML Models
Accessibility	Open Access
Purpose and relation to the objectives	To train AI models, test hypotheses, compare with newly generated data, and support conclusions
Data types	Text, Images, Numeric Data, Social Media Data
Data Formats	Plain Text, PDF, CSV, JSON, XML, Image Formats
Reuse of existing data	Yes, leveraging publicly available datasets

Data origin	Publicly available dataset or repository, previously generated by the project team, from previous research conducted by others
Expected size of the data	No more than 1PB
Data utility	The data will support AI model development and evaluation by enabling the training and testing of machine learning models to improve their trustworthiness, robustness against attacks like poisoning and backdoors, and protection mechanisms for models under continuous adaptation. Additionally, it will assist in evaluating explainability, fairness, bias, and the transparency of the models, while also assessing the effectiveness of protection strategies and the presence of biases in the data.
Privacy Principles	Whenever possible and needed, data will be open sourced with appropriate licensing, and its access made possible via public repositories. There is no plan to use protected or private data, but only publicly-available sources.

2.2.3 Dataset(s) for Context Awareness

Table 4: Dataset(s) for Context Awareness

Name of Dataset	Dataset(s) for Context Awareness
Accessibility	Open Access + Restricted Access for some datasets due to contractual, legal or ethical reasons
Purpose and relation to the objectives	To train AI models, test hypotheses, compare with newly generated data, and support conclusions
Data types	Text, Images, Audio, Numeric Data, Geospatial Data
Data Formats	Plain Text, PDF, CSV, JSON, XML, Image Formats, Audio Formats
Reuse of existing data	Yes, leveraging publicly available datasets
Data origin	Publicly available dataset or repository, from previous research conducted by others
Expected size of the data	A few TBs
Data utility	The data will support AI model training, features extraction, context-aware features development, performance testing and exploration of offline reinforcement learning approaches.
Privacy Principles	Data will be reviewed to ensure compliance with ethical guidelines (e.g., privacy and consent) and legal restrictions (e.g., licensing or intellectual property rights).

2.2.4 Dataset(s) for testing Anonymising Algorithms

Table 5: Dataset(s) for testing Anonymising Algorithms

Name of Dataset	Dataset(s) for testing Anonymising Algorithms

Accessibility	Open Access
Purpose and relation to the objectives	To test hypotheses, compare with newly generated data, and support conclusions
Data types	Text, Images, Numeric Data
Data Formats	Plain Text, PDF, CSV, JSON, XML, Image Formats
Reuse of existing data	Yes, leveraging publicly available datasets
Data origin	Publicly available dataset or repository, from previous research conducted by others
Expected size of the data	Several GBs
Data utility	The data will support the testing of anonymising algorithms capabilities.
Privacy Principles	Data will be reviewed to ensure compliance with ethical guidelines (e.g., privacy and consent) and legal restrictions (e.g., licensing or intellectual property rights).

2.2.5 Dataset(s) for Vulnerabilities Identification

Table 6: Dataset(s) for Vulnerabilities Identification

Name of Dataset	Dataset(s) for Vulnerabilities Identification
Accessibility	Restricted Access due to contractual, legal or ethical reasons
Purpose and relation to the objectives	To compare with newly generated data, and provide background information or context
Data types	Numeric Data, Vulnerability and Threat Assessment Data
Data Formats	JSON, XML, CVEs
Reuse of existing data	Yes, leveraging publicly available datasets
Data origin	Publicly available dataset or repository, from previous research conducted by others
Expected size of the data	Several GBs
Data utility	The data will be used in order to derive risk and security assessment measures.
Privacy Principles	No need to apply data aggregation, minimisation, and anonymisation methods to the data.

2.2.6 Dataset(s) for Pilot 1

Table 7: Dataset(s) for Pilot 1

Name of Dataset	Dataset(s) for pilot 1
Accessibility	Open Access + Restricted Access for some datasets due to contractual, legal or ethical reasons
Purpose and relation to the objectives	To test hypotheses and support conclusions
Data types	Images, Video, Numeric Data, Geospatial Data
Data Formats	CSV, Image Formats, Video Formats
Reuse of existing data	Yes, leveraging publicly available datasets

Data origin	Publicly available dataset or repository, Privately owned dataset or repository
Expected size of the data	Several GBs
Data utility	The data will serve as a basis for the use case 1 of pilot 1.
Privacy Principles	Data will be reviewed to ensure compliance with ethical guidelines (e.g., privacy and consent) and legal restrictions (e.g., licensing or intellectual property rights).

2.2.7 Dataset(s) for Pilot 2

Table 8: Dataset(s) for Pilot 2

Name of Dataset	Dataset(s) for Pilot 2
Accessibility	Open Access
Purpose and relation to the objectives	To test hypotheses, compare with newly generated data, and support conclusions
Data types	Text, Images, Video, Numeric Data, Geospatial Data
Data Formats	Plain Text, PDF, CSV, JSON, XML, Image Formats, Video Formats
Reuse of existing data	Yes, leveraging publicly available datasets
Data origin	Publicly available dataset or repository, previously generated by the project team, from previous research conducted by others
Expected size of the data	Less than 1 TB
Data utility	The data will be used for enabling injection of realistic LiDAR and GPS data poisoning attacks and for developing and evaluating robust AI models for resilient automotive perception. Additionally, it will be used to examine the impact of enhanced privacy-preserving FL frameworks in the aforementioned collaborative perception applications.
Privacy Principles	Data will be reviewed to ensure compliance with ethical guidelines (e.g., privacy and consent) and legal restrictions (e.g., licensing or intellectual property rights).

2.3 Administrative data

Administrative data are essential for managing the CoEvolution project. This includes reporting data, financial records, contact details, presentations, and other related materials. These data are mainly provided by project partners and exist in various formats. They are classified as confidential to protect proprietary and sensitive information. Table 9 outlines the formats, sources, and privacy considerations related to the administrative data.

Table 9: Administrative Data Description

Name of Dataset	Administrative Data
Accessibility	Restricted Access. Confidential Data. Data will be available only to CoEvolution Consortium Member and the EC and its representatives with granted access by the EC

Purpose and relation to the objectives of the project	Collect information for various purposes related to the management of the CoEvolution project
Data types	Unstructured (reporting data, financial data, contact information, presentations etc.)
Data Formats	DOC, PDF, XLS, PPT, CSV, TXT, HTML, JPG, TIFF, PNG, WEBM, MP4
Reuse of existing data	No reuse of existing data
Data origin	CoEvolution partners' contribution
Expected size of the data	GBs
Data utility	Useful for the project coordinator, partners' communication and collaboration, the EC.
Privacy principles	No need to apply data aggregation, minimisation, and anonymisation methods to the data.

2.4 Dissemination & Communication

Dissemination and communication data serve to enhance the outreach of the CoEvolution project. They contain details of various dissemination activities such as leaflets and videos, along with contact details and information about participation in events, workshops, and other project activities. Considering the sensitive nature of some data (e.g., contact details), access to such data is restricted as confidential, while materials for dissemination are open access. Table 10 provides more detailed characteristics of the dissemination and communication data.

Table 10: Dissemination & Communication Data Description

Name of Dataset	Dissemination & Communication Data
Accessibility	Materials used in dissemination activities, such as leaflets and videos, will be Open Access. However, data that is sensitive to privacy, such as contact details, will be treated as Confidential
Purpose and relation to the objectives of the project	To facilitate the communication and the operation of the project
Data types	Unstructured (Contact details of persons and other entities, participation details to events, workshops and other activities of the project, social media and other dissemination channels information and statistics etc.)
Data Formats	XLSX, DOC, MP4
Reuse of existing data	No reuse of existing data
Data origin	Partners, contacts of the project
Expected size of the data	GBs
Data utility	Data will be useful to project partners to carry out WP2 Tasks.
Privacy principles	No aggregation, anonymisation, or minimisation will be needed.

2.5 Exploitation

Exploitation data are paramount for the CoEvolution project to secure the project's sustainability and to facilitate the successful commercial application of its outcomes. Exploitation data encompass the details of the project's exploitation strategies and plans and are compiled primarily from the inputs of the project's partners and contacts. As this type of data often contains proprietary or business-sensitive information, it is classified as confidential to ensure appropriate protection. Table 11 further outlines the specifics of the exploitation data, including data formats, utility, and privacy considerations.

Table 11: Exploitation Data Description

Name of Dataset	Exploitation Data
Accessibility	Restricted Access. Confidential Data. Data will be available only to CoEvolution Consortium Member and the EC and its representatives with granted access by the EC
Purpose and relation to the objectives of the project	Exploitation plans and details
Data types	Unstructured
Data Formats	XLSX, DOC, PDF
Reuse of existing data	No reuse of existing data
Data origin	Partners, contacts of the project
Expected size of the data	Less than 1 GB
Data utility	Data will be useful to project partners to carry out WP2 Tasks.
Privacy principles	No aggregation, anonymisation, or minimisation will be needed.

2.6 Standardisation

Standardisation data represent an essential component of the CoEvolution project, as they reflect the project's alignment with existing standards and best practices. These data also facilitate the exchange of standardisation and feedback with standardisation bodies, policymakers, and other relevant parties, making them crucial for the project's adherence to the latest industrial standards. The data, generated by project partners and contacts, are provided as open access to facilitate their utility for project partners, standardisation organisations, and other relevant bodies. More details about the standardisation data are described in Table 12.

Table 12: Standardisation Data Description

Name of Dataset	Standardisation Data
Accessibility	Open Access
Purpose and relation to the objectives of the project	Alignment with standards and best practices across the framework. Exchange of standardisation and other feedback with standardisation bodies, policy makers and other parties
Data types	Unstructured
Data Formats	XLSX, DOC, PDF, TXT
Reuse of existing data	No reuse of existing data
Data origin	Partners, contacts of the project

Expected size of the data	Less than 1 GB
Data utility	Data will be useful to project partners, standardisation organisations, other relevant bodies.
Privacy principles	No aggregation, anonymisation, or minimisation will be needed.

3 FAIR Data

The management and use of data within the CoEvolution project are firmly guided by the FAIR data principles. The FAIR data principles represent a set of guidelines to enhance the ability of machines to automatically find and use data, in addition to supporting its reuse by individuals. The term “FAIR” is an acronym representing four foundational principles [1]:

- **Findability**: The data and supplementary materials have a unique and persistent identifier.
- **Accessibility**: Metadata and data should be easy to access. The protocol should be open, free, and universally implementable.
- **Interoperability**: The data should be structured such that it can be integrated with other data. Additionally, it should use a shared language for knowledge representation.
- **Reusability**: The data should be able to be reused for future research and have clear usage licenses.

As stated by the FORCE11 [2] group that developed these principles, “The FAIR Principles put specific emphasis on enhancing the ability of machines to automatically find and use the data, in addition to supporting its reuse by individuals” [3]. The implementation of these principles ensures that the data generated by the CoEvolution project can be effectively discovered, accessed, integrated, and reused, maximizing the project’s impact and efficiency.

Each dataset generated, used, or manipulated during the CoEvolution project will be treated in accordance with these principles, from data capture and storage, through to data analysis and sharing.

3.1 Making data findable, including provisions for metadata

Findability, as defined by the FAIR data principles, is central to the CoEvolution project’s data strategy. The project aims to ensure that all relevant datasets can be efficiently located, not only by project partners and external researchers but also by automated systems that rely on machine-readable metadata. To this end, the project will implement the following measures aligned with the FAIR sub-principles:

- **F1 - (Meta)data are assigned a globally unique and persistent identifier:** All datasets generated or collected by CoEvolution will be assigned globally unique and persistent identifiers, such as Digital Object Identifier (DOI) or Universally Unique Identifier (UUID), depending on the hosting repository. This will enable stable referencing and long-term traceability across project outputs.
- **F2 - Data are described with rich metadata (defined by R1 below):** The project will ensure that each dataset is accompanied by detailed metadata describing its context, structure, origin, and intended use. Metadata will include information such as data creators, collection date, methodology, versioning, and any applicable standards or protocols followed.
- **F3 - Metadata clearly and explicitly include the identifier of the data they describe:** CoEvolution will enforce consistent inclusion of dataset identifiers within metadata records. This direct reference will maintain integrity between the data and its description, supporting automated processing and linking.

- **F4 - (Meta)data are registered or indexed in a searchable resource:** All eligible datasets and metadata will be deposited in recognised repositories or catalogues (e.g., Zenodo), as specified in the Data Management Plan. This will ensure external visibility and compliance with Horizon Europe requirements. Internal datasets not publicly shared will still be indexed in a searchable internal repository accessible to project partners.

By following these steps, the CoEvolution project will make its data assets discoverable, traceable, and interoperable from the outset. This approach is essential for collaboration within the consortium, for transparency in reporting, and for enabling future reuse of the project's outputs.

3.1.1 Discoverability of the data

To bolster the discoverability of data within the CoEvolution project, we will implement a range of measures tailored to the classification of each dataset. These measures aim to ensure that data is readily accessible to authorised users.

CoEvolution underscores its commitment to effective data management in alignment with the FAIR data principles by tailoring our approach to enhance data discoverability while safeguarding confidentiality. These measures ensure that data remains accessible to authorised individuals and entities while upholding stringent security and privacy standards.

3.1.2 Data identification mechanisms

In open-access datasets, the focus will be on enhancing discoverability by incorporating search keywords and assigning unique DOI. This approach ensures that project members and relevant stakeholders can easily locate and reference these datasets. In cases where datasets are classified as confidential, meticulous attention will be given to enriching metadata manually. This metadata will provide comprehensive insights into the data's nature, origin, and relevance. However, access to such rich metadata will be restricted to approved users to maintain the confidentiality and security of sensitive data.

3.1.3 Naming conventions used

Adopting a systematic and coherent naming convention for data files is essential for ensuring efficient organisation and retrieval. The naming convention adopted will encompass clear, concise, and meaningful identifiers that reflect the nature and context of the data. For instance, a typical naming format can be the following:

"CoEvolution_[DataCategory]_[DataSetType]_[YearMonth]".

This format indicates the project name and provides immediate insight into the data type and the timeframe of its creation or update.

3.1.4 Clear versioning of the documents

Clear document versioning will be maintained throughout the project to ensure up-to-date data is readily available. This approach will be standardized across all datasets to ensure data accuracy. The process will involve:

- Version Numbering
- Modification Date
- Change Log Maintenance
- Standardisation Across Datasets

This structured approach to document versioning aligns with the best practices recommended. It is pivotal in maintaining the integrity and utility of the data throughout the project's lifecycle.

3.1.5 Standards for metadata creation

In the CoEvolution project, the standards for metadata creation will be diligently aligned with recognised international standards tailored to the specific nature of each dataset. For instance:

- **ISO/IEC 11179 Standard:** This standard, known for its comprehensive approach to metadata registry, will be employed for specific datasets where detailed, structured metadata is crucial. For example, an "Infrastructure Security Analysis" dataset could follow this standard to ensure metadata is robust and systematically organised.
- **Asset Administration Shell (AAS) Standard:** The AAS standard will be utilised for datasets requiring a more manually intensive approach to metadata creation. This is particularly relevant for datasets with complex or concrete data structures.

This strategic approach to metadata creation ensures that all datasets within the CoEvolution project are accompanied by high-quality, standard-compliant metadata, facilitating effective data management and usability.

3.2 Making data openly accessible

The second principle of the FAIR data principles is Accessibility. Accessibility is about ensuring that once data is found, it can be accessed by both humans and machines. There are two sub-principles related to this:

- **A1 - (Meta)data are retrievable by their identifier using a standardized communications protocol:** This principle asserts that data should be retrievable using the unique identifier assigned to it (referenced in F1) through a standardized communications protocol. A standardized protocol means a commonly accepted and widely used method of communication between systems. This could be as simple as a HTTP/HTTPS protocol (the standard protocol for data transmission over the internet), or it might involve more specialized data transfer protocols depending on the nature of the data. The key is that this protocol should be universally recognized and adopted, so that data retrieval can be automated and can occur without the need for specialized software or tools.
  - **A1.1 - The protocol is open, free, and universally implementable:** This is a clarification of the previous point. Not only should the protocol be standardized, but it should also be open, free, and universally implementable. This means that it is available for anyone to use, without any restrictions or fees. This is important for promoting wide usage and accessibility of the data.

  - **A1.2 - The protocol allows for an authentication and authorization procedure, where necessary:** In cases where data access needs to be controlled or restricted (for example, for privacy reasons), the protocol should support authentication and authorization. Authentication is the process of verifying the identity of the user requesting access, while authorization is about determining what level of access that authenticated user should have.

- **A2 - Metadata are accessible, even when the data is no longer available:** This principle ensures that even if the actual data becomes unavailable (for instance, if it's been removed, or if access to it has been restricted), the metadata remains accessible. The rationale for this is that even without the actual data, the metadata can still provide valuable information. It can support the understanding of what kind of data was collected and how, helping future research projects in their design or in the understanding of the evolution of a particular research field.

Accessibility, like Findability, is crucial in the context of the FAIR data principles, and in the broader context of data management, as it ensures that data can be used and reused effectively and efficiently.

3.2.1 Methods or software needed to access the data

Methods to access data will differ based on data classification. For the Open Access Datasets, access will be granted through a standardised protocol like an Application Programming Interface (API) or web services. For Datasets of restricted or confidential nature, a user registration process and password protection protocols will be enforced.

3.2.2 Deposit of data, associated metadata, documentation and code

This section focuses on the protocols for depositing data, associated metadata, documentation, and code. The deposition will be tailored to suit the nature of each dataset, ensuring secure and efficient storage.

Within the CoEvolution project, a structured and secure process will be established to deposit data, associated metadata, documentation, and code. This process aligns with the project's commitment to the best practices of FAIR data principles and data management.

Trusted Repositories and Private Storage: Depending on the dataset's sensitivity and requirements, storage will either be in trusted repositories or private infrastructure. For instance, specific datasets may be stored in specialised repositories provided by the project's research partners or private infrastructures tailored to their data type provided by the pilot partners.

Examples: A sensitive cybersecurity threat analysis dataset might be stored in a private, secure infrastructure, whereas a dataset on public cybersecurity awareness could be deposited in an open-access repository for broader reach.

3.3 Making data interoperable

For data to be fully effective, they must be interoperable, meaning they should be designed and structured in a way that allows them to be integrated, combined, and analyzed with other datasets. In the context of the FAIR data principles, interoperability is dissected into three key aspects:

- **I1. (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation:** it is critical that the data and metadata are represented using a standardized language that is broadly understood and widely accepted. This can make it much easier to integrate data from different sources or to use different types of data together.
- **I2. (Meta)data use vocabularies that follow the FAIR principles:** Using standard vocabularies that are FAIR-compliant further enhances interoperability. When data from different sources use the same vocabularies, it becomes much easier to integrate and analyze that data in a unified way.
- **I3. (Meta)data include qualified references to other (meta)data:** To be interoperable, data should not only be able to stand alone but also fit into a broader context. This is achieved by providing qualified references to other related data or metadata.

3.3.1 Interoperability of data assessment

To ensure data interoperability, we will follow identified data and metadata standards and best practices. Documentation will be provided, including for example ReadMe files and detailed documentation of data provenance, to facilitate data interoperability.

3.3.2 Vocabulary use

The specific vocabulary and ontologies used will be project-specific or based on common ones in the field and will be determined as per the nature of the dataset. Consultations with field experts may also be considered for uncommon ontologies or vocabularies.

3.4 Making data re-usable

Reusability is the last of the FAIR principles, emphasizing the importance of ensuring that data and metadata are not just a one-time resource but have the potential for further applications.

The principle of reusability can be broken down into four key components.

- **R1. (Meta)data needs to be richly described with a plurality of accurate and relevant attributes:** Rich, accurate, and relevant descriptions of data and metadata enhance their reusability. By providing extensive details about the data, potential users can better understand its context, quality, and applicability for their own research.
  - **R1.1. (Meta)data should be released with a clear and accessible data usage license:** A clear and accessible data usage license is essential for reusability. It informs potential users about their rights and obligations regarding the use of the data.
  - **R1.2. (Meta)data should be associated with detailed provenance:** Detailed provenance information adds to the reusability of data by providing context about

how the data was collected, processed, and analyzed. This allows potential users to assess the quality and reliability of the data.

  – **R1.3. (Meta)data should meet domain-relevant community standards:** Meeting community standards improves the reusability of data by ensuring it aligns with what other researchers in the same field expect and understand.

In conclusion, the principle of reusability encourages us to treat our data as a valuable resource with a life cycle extending beyond the immediate project. It compels us to strive for transparency, rigor, and alignment with community norms, thus maximizing the potential of our data to contribute to further research and innovation.

3.4.1 Increase data re-use through clarifying licenses

As part of the project's DMP, appropriate licenses are applied to enhance data reuse and research outputs. All datasets and research outputs in CoEvolution will be accompanied by clear licenses to encourage free reuse and adaptation. Below are some indicative examples:

- **Example for Software and Models:** Outputs such as software or AI models may be licensed under the Creative Commons Attribution 4.0 International License, allowing free use and adaptation with proper attribution and sharing derivatives under the same terms.
- **Example for Data Sharing:** Datasets like real-time cyber-attack patterns might be licensed under the GNU General Public License to promote open collaboration and ensure that derivative works are freely available, thus fostering a collaborative ecosystem in cybersecurity research.

This approach maximises the impact and utility of CoEvolution's outputs by facilitating broad and efficient reuse in the research and innovation community.

3.4.2 Data quality assurance process

In the CoEvolution project, data quality and integrity are critical to achieving their research objectives. A comprehensive data quality assurance process has been established to ensure the data is accurate, reliable, and high-quality. This process encompasses a range of validation checks, error detection routines, and manual data review and verification procedures.

3.4.3 Length of time for which the data will remain re-usable

While the specific timeframe may vary, data will be preserved for a minimum duration to ensure long-term accessibility and usability. This will be facilitated through reliable repositories and regular data updates.

3.5 Data description templates in CoEvolution

Within the CoEvolution project, data will be captured and processed following the FAIR principles to ensure the highest level of Findability, Accessibility, Interoperability, and Reusability. To facilitate this process, we have developed a series of data description templates. These templates are designed to assist the project partners in systematically capturing, documenting, and sharing data generated during the project.

These data description templates will cover a range of data types relevant to the CoEvolution project, ensuring that all data, irrespective of its format or origin, is treated with the same high standards of data management. The templates will guide the data providers in capturing the necessary information to ensure compliance with the FAIR principles. They will help establish clear and consistent metadata, ensure appropriate licensing, document detailed data provenance, and align the data with domain-relevant community standards.

Table 13: Making Data Findable

Making Data Findable	
Name of data set	Univocal identifier of the considered data: [e.g. CoEvolution_WPx_Tx.z_001].
Data types	Describes the nature of the data (e.g., real-time data stream, unstructured like tweets, synthetic data stream).
Data generation and/or collection	Description of the type of input used to generate the data and the complete methodology and tools used for data collection.
Purpose	What are the data collected/generated specifically used for.
Data origin	Where applicable, information from applications to be developed by the partner.

Table 14: Making Data Accessible

Making Data Accessible	
Accessibility	Open/Restricted.
Repository	Description/location of the available data.
Shareability restrictions/related Information	Where applicable, information from applications to be developed by the partner.

Table 15: Making Data Interoperable

Making Data Interoperable	
Format	Data format, measuring unit, typical order of magnitude.
Expected size of the data	An estimation, e.g., 1 TB/Day or 8 GB/day when compressed.
Standards and metadata	The metadata attributes list. The used methodologies.
Standard software Interfaces	List of the standards used to promote results replicability.
Extensions to standard interfaces	Extensions to the above standards as developed during the project.

Table 16: Making Data Reusable

Making Data Reusable	
RE-use of existing data	Information on the reuse of existing data, if applicable.
Data backup	Consistent location of the data, including previous releases.
Quality Consistency	Constraints determining the quality/currency of the collected data.

Emulation tools	Description/location of possible emulation tools useful for replicating the data.

4 Allocation of Resources

4.1 Data management responsibilities

Data will be stored in the Collaboration file repository (Microsoft Sharepoint), set by the Project Coordinator (PC) as the project repository, and will be kept at least for 6 months after the end of the project. Extended storage for additional years will be granted upon request. CoEvolution's Ethic Advisor (AEGIS) and PC (ICCS), assisted by the WP leaders, will jointly manage all data-related responsibilities. These responsibilities include the following:

- the creation and regular updating of the DMP
- ensuring suitable storage, management, and sharing of data
- establishing a strategy for efficient data use and high-quality storage.

With respect to the Datasets provided by the Pilots and Technical partners, data management responsibilities will be shared across CoEvolution partners, each overseeing their own datasets. In the cases that will be deemed necessary a Data Protection Officer (DPO) will be appointed by the corresponding partner.

4.2 Cost of potential value of long-term preservation

The project's funds will cover the essential costs of adhering to FAIR data principles, including data publication, updating, maintenance, and security. Costs related to article publishing in Open Access journals are also addressed as per Article 6 and ANNEX 5 of the Grant Agreement (GA).

It is challenging to provide an exact estimate for long-term data preservation costs post-project in this version of the DMP. Preservation on open repositories such as Zenodo, which offer up to 50 GB free storage per dataset, and ICCS's private Microsoft Sharepoint platform, will be leveraged as cost-effective solutions. However, the final decision on long-term preservation costs will be considered based on future necessities.

5 Data Security

The CoEvolution consortium is committed to protecting the confidentiality, integrity, and availability of all personal and research data handled throughout the project. Each partner is responsible for ensuring that data processing complies with the GDPR and any other applicable legal or ethical requirements.

5.1 Data storage and access management

The project will use a secure instance of Microsoft SharePoint Online, hosted within the Microsoft 365 environment, as its primary platform for storing and managing project data. All information will be encrypted in transit using Transport Layer Security (TLS) and at rest using AES 256-bit encryption, in accordance with recognised industry standards.

Access to data will be restricted based on clearly defined roles and responsibilities, following the principle of least privilege. Administrative privileges—such as file deletion or access management—will be limited to designated personnel. Multi-Factor Authentication (MFA) will be required for all user accounts.

SharePoint's built-in access management capabilities will allow for granular control over who can view or modify specific data. In addition, Microsoft 365's enterprise-level infrastructure offers robust protection through tools such as audit logging (Microsoft Purview), threat monitoring (Microsoft Defender), and automatic versioning, which supports recovery of previous file states. Deleted files can be recovered during a configurable retention period, adding further resilience.

All Microsoft 365 services used in the project operate under the EU Data Boundary initiative. This means that all personal data—including backups and geo-redundant storage—will be stored and processed exclusively within the European Union, ensuring full compliance with GDPR requirements on data localisation and sovereignty. Microsoft does not access or process customer data beyond what is necessary to deliver its services and complies with standards such as ISO/IEC 27001.

In addition to SharePoint, some partners may maintain local or institutional copies of project data. These systems must implement equivalent safeguards, including full-disk encryption, strong password protection, regular updates, antivirus and anti-malware protection, and compliance with institutional firewall and network policies. Where possible, sensitive data will not be stored locally, unless required for specific project tasks.

Together, these technical and organisational measures will ensure that project data is handled securely, ethically, and in full compliance with legal and institutional data protection requirements.

5.2 Data sharing and transfers

When data must be exchanged between partners, transfers will occur through secure, encrypted channels (e.g., encrypted SharePoint links, secure email, or Microsoft Teams). Prior to sharing, personal or sensitive data will undergo anonymisation or pseudonymisation whenever possible, in accordance with the principle of data minimisation outlined in the GDPR.

Only the essential amount of data needed to fulfill the project's objectives will be shared. Appropriate safeguards will be maintained throughout each transfer to preserve data confidentiality and integrity.

5.3 Data backup and recovery

To ensure data resilience, SharePoint provides integrated backup and restoration features. Deleted files are retained for a configurable period via the platform's Recycle Bin, and version history is maintained to allow recovery of previous document states if needed.

In addition, Microsoft's infrastructure includes geo-redundant backup systems, ensuring that data is stored in multiple secure data centers within the EU. This reduces the risk of loss in the event of hardware failure or service interruption.

Partners who maintain local copies of project data are expected to implement similar backup routines, enabling them to restore critical information in the event of accidental deletion, hardware malfunction, or other disruptions.

5.4 Response to data incidents

The project fully recognizes its responsibilities under Articles 33 and 34 of the GDPR, which require prompt action in the event of a personal data breach. All partners must take immediate steps to:

- Limit and contain any potential breach
- Assist in determining the scale and root cause of the incident
- Support the notification of data protection authorities and affected individuals, where applicable

All partners are expected to act swiftly and cooperatively in handling any suspected or confirmed breaches. Each partner shares the duty to uphold GDPR obligations and maintain transparency in addressing security incidents.

Through the use of Microsoft SharePoint's secure infrastructure, strict access control practices, encryption, backup systems, and a clear commitment to GDPR compliance, the CoEvolution project maintains a high standard of data protection. These measures collectively ensure that all personal and research data is treated with care and responsibility, throughout the project's duration and beyond.

6 Ethics and Legal Aspects

The ethical and legal aspects related to the project are developed and addressed in the context of a specific working group (WP1). The CoEvolution partners are fully committed to upholding ethical standards and research integrity throughout the duration of the project, in accordance with the provisions set forth in the GA. Furthermore, all activities will be conducted in alignment with applicable human rights legislation, including the EU Charter of Fundamental Rights, the European Convention on Human Rights, the Non-Discrimination European legislation, and other relevant European and international legal instruments. This commitment reflects a broader dedication to ensuring that the development and implementation of the Project remain consistent with core European values and the protection of fundamental human rights.

CoEvolution activities are expected to utilise both real and synthetic data. The synthetic data, mimicking the characteristics and patterns of real-world data, are derived from real data, but are not real (actual personal data). Synthetic data is gaining ground because of its advantages in terms of privacy preservation and is considered as Privacy Enhancing Technology (PET) as the data used in the training process is not directly addressed to an identified or identifiable person [4].

However, both real and synthetic data are not without privacy risks. While synthetic data is not inherently private [5] and may be vulnerable to privacy attacks and breaches that could lead to the disclosure of the actual data from which it was derived through re-identification, real data also require careful handling to avoid privacy concerns. Therefore, even when synthetic data is used, as well as when real data are incorporated, vigilance and proactive measures are necessary to manage risks and to ensure privacy guarantees. Regular audits and tests will help to identify any vulnerabilities in the systems to potential privacy attacks and to ensure that appropriate safeguards are in place to preserve data privacy.

Real data may contain inherent biases, but since synthetic datasets are derived from existing real datasets, it is likely to incorporate or even reinforce existing biases present in them. To address these challenges, mechanisms such as bias detection algorithms, bias reduction techniques and fairness-aware data generation techniques, can be implemented to detect, reduce or eliminate any biases.

Privacy preservation will be tested through privacy assurance assessment following the guidelines of the EDPS to ensure that the projects' generated synthetic data is not actual personal data. The Privacy Assurance Assessment will evaluate the degree to which data subjects can be identified in the synthetic data, as well as the extent of additional information that could be inferred about them upon successful identification [6].

Although it is not expected that personal data will be collected under CoEvolution, should this occur, partners will fully comply with relevant laws and regulations of European legislation on the processing of such data, particularly the GDPR [7], by ensuring that such data is handled in accordance with its provisions.

CoEvolution will implement all necessary measures to protect the anonymity of data that are shared among the partners. Each party that discloses or otherwise makes shared information available represents and warrants that it possesses the requisite authority to do so, and that no legal, contractual, or other restrictions exist which would preclude any other party from utilizing such shared information for the purposes of this action and its subsequent exploitation.

Furthermore, a group of internal and external advisors, will monitor partners' compliance with applicable European laws and regulations and provide guidance on issues that may raise concerns regarding ethical values and the protection of human rights, including the protection of personal data. This will be documented in the deliverable D1.4 "Analysis of the legal and ethical issues" at M12

6.1 Compliance with GDPR and other legislation and regulations

The CoEvolution partners are fully aware of the existing legal framework, as well as the obligations and requirements it sets forth when personal data is collected and processed. The Consortium is committed to complying with the GDPR and fully implementing its provisions. The seven fundamental principles outlined in the GDPR—lawfulness, fairness, and transparency; accuracy; purpose limitation; data minimization; integrity and confidentiality; storage limitations; and accountability—will be strictly observed when personal data is collected and processed.

Furthermore, prior to the collection and processing of personal data, explicit written consent shall be obtained from the data subjects. Individuals will be fully informed of their rights concerning their personal data, as defined under the GDPR, as well as the procedures for exercising those rights. In cases where the transfer of personal data to countries outside the European Union is necessary for the implementation of the project, CoEvolution will ensure that such transfers are carried out in compliance with GDPR requirements, specifically by relying on an adequacy decision issued by the European Commission to guarantee an equivalent level of data protection.

The consortium undertakes not to use personal data for commercial purposes and not to disclose it to third parties, maintaining the confidentiality of all shared information. The personal data collected for the registration of the user on the platform will be processed in accordance with the privacy policy that will provide the user with all the necessary information in accordance with the GDPR, including who is responsible for the processing of his/her data and the rights he/she has in relation to the personal data.

In addition to the GDPR, CoEvolution is also committed to comply with all relevant existing European and international legislation for the protection of human rights, in particular:

- The Charter of Fundamental Rights of the European Union [8]
- The Convention for the Protection of Human Rights and Fundamental Freedoms [9]
- The Regulation (EU) 2019/881 of the European Parliament and of the Council of 17 April 2019 on ENISA (the European Union Agency for Cybersecurity) [10]
- Directive (EU) 2022/2555 of the European Parliament and of the Council of 14 December 2022 on measures for a high common level of cybersecurity across the Union (NIS2) [11]

CoEvolution's compliance with all the above European laws and with relevant national, European, and international legislation and regulations, as well as soft laws and documents related to data handling and protection, will ensure that the project's activities are aligned with the highest legal and ethical standards.

6.2 Use of A.I.

CoEvolution recognizes the prominent role that AI ACT has in regulating AI to protect fundamental rights and is committed to fully implement the principles and requirements of the Act to ensure trustworthy AI during the project and after its completion. Emphasis will be placed on aligning the project with legislative requirements for transparency, human oversight, safety and accountability.

At this stage, the use of high-risk AI tools and systems for the project's activities is not anticipated, in line with the risk-based classification set out in the AI Act. However, ongoing monitoring and assessment will be carried out throughout the project's implementation to ensure that any AI tools and systems employed fully comply with the obligations and requirements established by the Act.

Furthermore, CoEvolution will use self-assessment tools, such as the Assessment List for Trustworthy Artificial Intelligence (ALTAI), to ensure that the AI tools and systems employed are trustworthy and aligned with fundamental ethical principles and requirements. The application of these tools will be crucial in supporting the project's adherence to the Ethics Guidelines for Trustworthy AI [12] by enabling the early identification and mitigation of ethical concerns and potential risks associated with AI systems, while promoting alignment with the principles of transparency and accountability.

7 Conclusions

The CoEvolution DMP provides an organized and detailed framework for handling, processing, and preserving data throughout the project's lifecycle. It is the cornerstone of our commitment to maintaining high standards of data management, ensuring optimal usage of data, and promoting an environment of transparency and collaboration. The document outlined the types of data we will generate, collect, and process in the context of the CoEvolution project, the methodologies and standards applied, and the ways we plan to preserve and share the data. We highlighted our commitment to Open Access publishing, which will ensure that our findings can be widely shared, fostering scientific collaboration and progress.

We have also addressed the significance of ethical considerations, legal aspects, and compliance with the GDPR and other data protection legislation in our work. These measures will ensure the protection of any data involved in the project, maintain trust, and align our work with the best ethical practices and legal requirements. The use of AI in CoEvolution serves a crucial role. Our approach to AI integrates compliance with the AI Act, thus ensuring safe, transparent, and law-respecting AI applications.

The DMP is a living document, and it will be reviewed and updated periodically to reflect any changes or advances in the CoEvolution project. There are two scheduled updates at M12 and M18 to capture the planned developments and any new requirements that may have been identified. Moreover, interim updates will be made whenever significant changes occur in the project. Each iteration will build upon the existing data management best practices and accommodate any newly identified requirements or advancements in data management strategies.

In conclusion, the CoEvolution project is committed to exemplary data management, promoting transparency, collaboration, and scientific excellence. Through the implementation of this DMP we contribute not only to the success of the project, but also to the building of high standards for data integrity, security, and accessibility.

References

[1] GO FAIR Initiative, "Fair principles," accessed: 2025-04-27. [Online]. Available: https://www.go-fair.org/fair-principles/

[2] Force11, "Force11," accessed: 2025-04-27. [Online]. Available: https://force11.org/

[3] M. D. Wilkinson *et al.*, "The fair guiding principles for scientific data management and stewardship," *Scientific Data*, vol. 3, 2016.

[4] European Data Protection Supervisor, "Techsonar report 2021-2022," dec 2021, last accessed: 2025-04-14. [Online]. Available: https://www.edps.europa.eu/system/files/2021-12/techsonar_2021-2022_report_en.pdf

[5] J. Hradec, M. Craglia, M. Di Leo, S. De Nigris, N. Ostlaender, and N. Nicholson, "Multi-purpose synthetic population for policy applications," Publications Office of the European Union, Luxembourg, Tech. Rep. EUR 31116 EN, 2022, jRC128595.

[6] European Data Protection Supervisor, "Synthetic data," last accessed: 2025-04-14. [Online]. Available: https://www.edps.europa.eu/press-publications/publications/techsonar/synthetic-data_en

[7] European Parliament and Council, "Regulation (eu) 2016/679 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (general data protection regulation)," 2016, last accessed: 2025-04-14. [Online]. Available: https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng

[8] European Parliament, "Charter of fundamental rights of the european union (2000/c 364/01)," last accessed: 2025-04-14. [Online]. Available: https://www.europarl.europa.eu/charter/pdf/text_en.pdf

[9] European Court of Human Rights, "Convention for the protection of human rights and fundamental freedoms," last accessed: 2025-04-14. [Online]. Available: https://www.echr.coe.int/documents/d/echr/convention_ENG

[10] European Parliament and Council, "Regulation (eu) 2019/881 on enisa (the european union agency for cybersecurity) and on information and communications technology cybersecurity certification and repealing regulation (eu) no 526/2013 (cybersecurity act)," 2019, last accessed: 2025-04-14. [Online]. Available: https://eur-lex.europa.eu/eli/reg/2019/881/oj/eng

[11] E. Parliament and Council, "Directive (eu) 2022/2555 on measures for a high common level of cybersecurity across the union, amending regulation (eu) no 910/2014 and directive (eu) 2018/1972, and repealing directive (eu) 2016/1148 (nis 2 directive)," 2022, last accessed: 2025-04-14. [Online]. Available: https://eur-lex.europa.eu/eli/dir/2022/2555/oj/eng

[12] E. Commission, C. Directorate-General for Communications Networks, Technology, and G. ekspertów wysokiego szczebla ds. sztucznej inteligencji, *Ethics guidelines for trustworthy AI.* Publications Office, 2019.

Appendix A
CoEvolution Data Management Plan Questionnaire v1.0

In Appendix A, it is included the Questionnaire (v1.0) completed by each project partner, which served as a foundational input for this deliverable. The responses provided insights into current data management practices, anticipated data types, storage solutions, sharing protocols, etc.

CoEvolution

A Comprehensive Trustworthy Framework for Connected Machine Learning and Secure Interconnected AI Solutions.

Data Management Plan Questionnaire

CoEvolution has received funding from the European Union's Horizon Europe research and innovation programme under Grant Agreement No. 101168560.

Table of Contents

1 Introduction

The Horizon Europe Model Grant Agreement requires a Data Management Plan (DMP) to be established and regularly updated. In the CoEvolution project, this requirement will be addressed via the creation and submission of the deliverable D1.2: Data Management Plan

The following questionnaire aims to serve as the primary input source for creating the DMP. Therefore, all partners are kindly requested to complete the questionnaire and provide their input. Following the structure of the template that HORIZON EUROPE has recently released, the questionnaire consists of the following main sections:

- **"Data Summary"** focuses on questions about any existing data to be reused, types and formats of data generated, the purpose and relation to project objectives, expected data size, origin, and potential utility for external parties.
- **"FAIR Data"** includes questions aiming to address making data findable, accessible, interoperable, and reusable by discussing persistent identifiers, metadata provisions, repository arrangements, data accessibility and restrictions, interoperability standards, and measures to increase data re-use while considering resource allocation, data security, and ethical aspects.
- **"Other Research Outputs"** section emphasises questions related to managing and planning for both digital and physical research outputs generated or reused throughout the project and applying FAIR data principles to these.
- **"Allocation of Resources"** delves into the costs of making data and research outputs FAIR, covering direct and indirect expenses, funding sources, the assignment of data management responsibilities, and the resources required for long-term preservation and data retention decisions.
- **"Data Security"** enlists questions that address the measures implemented for data security, including data recovery, secure storage, archiving, and transfer of sensitive data, along with the assurance of safe storage in trusted repositories for long-term preservation and curation.
- "**Ethics"** aims to explore potential ethical and legal issues impacting data sharing, discussing informed consent for data sharing and long-term preservation concerning personal data and referencing relevant ethics deliverables and chapters.
- "**Personal Data"** enlists questions related to the Use and Management of personal data.
- "**Artificial Intelligence"** dives into the possible use of A.I. technologies for data management.
- "**Other Issues"** examine the use of national, funder, sectorial, or departmental procedures for data management, listing and briefly describing the chosen procedures.

2 Partner Information

Partner	
Corresponding person (email)	
Name of the Dataset	*Please Specify the name of the dataset (if applicable)*
Name of the Use Case	*Please Specify the use case that this dataset will be used (if applicable)*

3 Data Summary

1. Will you be re-using any existing data in this project?

☐ Yes ☐ No

2. If you answered "yes" to the previous question, what will you re-use the existing data for? (Select all that apply)

☐ To test hypotheses
☐ To support conclusions
☐ To compare with newly generated data
☐ To provide background information or context
☐ n/a
☐ Other (Please Specify) *Click or tap here to type your answer.*

3. What is the source of the existing data that you intend to re-use? (Select all that apply)

☐ Publicly available dataset or repository
☐ Privately owned dataset or repository
☐ Previously generated by the project team
☐ From previous research conducted by others
☐ Other (Please specify): *Click or tap here to type your answer.*

4. What criteria did you use to determine the suitability of the existing data for re-use? (Select all that apply)

☐ Data quality
☐ Data format and compatibility
☐ Data relevance to the project's objectives
☐ Availability of metadata
☐ Compliance with legal and ethical requirements
☐ Other (Please specify): *Click or tap here to type your answer.*

5. Have you obtained permission or authorization from the original data owners to re-use their data?

☐ Yes ☐ No ☐ Other (Please specify):*Click or tap here to type your answer.*

6. Are there any limitations or restrictions to the use of the existing data, such as copyright or confidentiality agreements?

CoEvolution has received funding from the European Union's Horizon Europe research and innovation programme under Grant Agreement No. 101168560.

☐ Yes No
7. What types of data will be collected/ generated or re-used in the project? (Select all that apply)
☐ Text ☐ Images ☐ Audio ☐ Video ☐ Numeric data (e.g., measurements, statistics) ☐ Geospatial data (e.g., maps, GPS coordinates) ☐ Genetic data ☐ Survey or questionnaire data ☐ Social media data ☐ Other (please specify):*Click or tap here to type your answer.*
8. In what formats will the data be collected/ generated or re-used? (Select all that apply)
☐ Plain text ☐ PDF ☐ CSV ☐ JSON ☐ XML ☐ RDF ☐ Image formats (JPEG, PNG, etc.) ☐ Audio formats (MP3, WAV, etc.) ☐ Video formats (MP4, AVI, etc.) ☐ Other (please specify):*Click or tap here to type your answer.*
9. Will the data be structured or unstructured?
☐ Structured (data that has a defined schema or format) ☐ Unstructured (data that does not have a defined schema or format) ☐ Both structured and unstructured ☐ Not sure at this stage
10. If the data is structured, what will be its structure? (Select all that apply)
☐ Spreadsheet (e.g., Excel, Google Sheets) ☐ Hierarchical structure (e.g., XML, JSON) ☐ Graph structure (e.g., RDF, OWL) ☐ Other (Please specify): *Click or tap here to type your answer.*
11. In what types of storage formats will the data be stored? (Select all that apply)
☐ Relational database (e.g., SQL) ☐ NoSQL database (e.g., MongoDB) ☐ Spreadsheet (e.g., Excel, Google Sheets) ☐ Text file (e.g., CSV, TSV) ☐ Binary file (e.g., image, audio, video) ☐ Cloud storage (e.g., AWS S3, Google Cloud Storage) ☐ Other (please specify): *Click or tap here to type your answer.*
12. What is the purpose of the data collection/ generation or re-use and its relation to the objectives of the project?

Click or tap here to type your answer.
13. What specific research questions will the data help to answer?
Click or tap here to type your answer.
14. Can you estimate the expected volume of data that will be collected/ generated or re-used in the project?
Click or tap here to type your answer.
15. Will the data be collected/ generated over a period of time, and if so, what is the expected rate of data collection/ generation?
Click or tap here to type your answer.
16. How will the data be managed and stored to ensure that it remains accessible and usable over time? (Select all that apply)
☐ Data will be stored in a trusted repository. ☐ Data will be stored in multiple locations to prevent loss. ☐ Data will be stored in an encrypted format. ☐ Data will be stored in a cloud-based storage service. ☐ Data will be regularly checked for errors and inconsistencies. ☐ Data will be regularly tested for security vulnerabilities. ☐ Data will be regularly audited for compliance with ethical and legal standards. ☐ Data will be backed up regularly. ☐ Data will be made available in open and standardized formats. ☐ Data will be assigned persistent identifiers to ensure findability. ☐ Access to the data will be controlled through a formal access policy. ☐ Data will be preserved for a minimum number of years. ☐ Data will be regularly refreshed or updated to maintain its usefulness. ☐ Data will be versioned to track changes over time. ☐ Data will be migrated to new formats if necessary. ☐ Data will be anonymized or de-identified if necessary. ☐ Other (please specify):*Click or tap here to type your answer.*
17. Can you describe the origin/provenance of the data, either collected/ generated or re-used including how it was collected/ generated and any associated metadata?
Click or tap here to type your answer.
18. Are there any quality issues with the data, such as missing values or errors, and if so, how will these be addressed?
Click or tap here to type your answer.
19. Is the data representative of the topic or phenomenon of interest?
☐ Yes ☐ No
20. Who are the potential beneficiaries of the data that you will collect/ generate or re-use? (Select all that apply)

☐ Other researchers in the same field
☐ Researchers in other fields
☐ Industry partners or companies
☐ Government agencies or policymakers
☐ Non-governmental organizations or advocacy groups
☐ The general public
☐ Educators or students
☐ Other (please specify):*Click or tap here to type your answer.*

21. What is the potential impact of the data on the wider research community, industry, or society? (Select all that apply)

☐ Advancing scientific knowledge and discovery
☐ Driving innovation and economic growth
☐ Informing policy and decision-making
☐ Improving public health and well-being
☐ Enhancing education and training
☐ Enabling new technologies or applications
☐ Fostering international collaboration and partnerships
☐ Other (please specify):*Click or tap here to type your answer.*

22. How will you ensure that the data is discoverable and accessible to others who might be interested in using it? (Select all that apply)

☐ Depositing data in a trusted repository
☐ Providing metadata to enable discovery.
☐ Assigning persistent identifiers to the data
☐ Making data available through a standardized access protocol
☐ Enabling open access to the data
☐ Offering search keywords to optimize discovery.
☐ Establishing a data access committee to evaluate and approve access requests.
☐ Providing documentation to validate data analysis and facilitate re-use.
☐ Other (please specify):*Click or tap here to type your answer.*

4 FAIR DATA

4.1 FAIR DATA - Making data findable, including provisions for metadata.

23. Will you assign a unique identifier to the data to ensure it can be consistently identified and cited?
☐ Yes ☐ No
24. Which identifier scheme(s) will you use for your data? (Select all that apply)
☐ DOI ☐ URN ☐ ARK ☐ PURL ☐Other (Please Specify) *Click or tap here to type your answer.*
25. Will rich metadata be provided to allow discovery?
☐ Yes ☐ No
26. What metadata will be created? (Select all that apply)
☐ Descriptive metadata (e.g., title, author, date, description) ☐ Administrative metadata (e.g., access restrictions, provenance, file format) ☐ Structural metadata (e.g., relationships between different data objects) ☐ Technical metadata (e.g., file size, resolution, software used) ☐ Other (Please specify): *Click or tap here to enter text.*
27. What disciplinary or general standards will be followed? (Select all that apply)
☐ Dublin Core ☐ Darwin Core ☐ DataCite Metadata Schema ☐ ISO/IEC 11179 ☐ FGDC Metadata Standard ☐ EML (Ecological Metadata Language) ☐ Other (Please specify):*Click or tap here to enter text.*
28. Will the metadata be created manually or using automated tools?
☐ Manually ☐ Automated ☐ Both ☐ Not Applicable ☐ Other (Please specify):*Click or tap here to enter text.*
29. If metadata standards do not exist in your discipline, what type of metadata will be created and how will it be structured to enable discovery and interoperability?
Click or tap here to type your answer.
30. Will search keywords or tags be provided in the metadata to optimize the possibility for discovery and then potential re-use?

☐ Yes ☐ No
31. How will you select the keywords?
☐ Manually ☐ Using automated tools ☐ Combination of both. ☐ Other (Please specify): *Click or tap here to enter text.*
32. Will the keywords be standardized across the project or left to individual researchers to assign?
☐ Standardized across the project. ☐ Left to individual researchers to assign. ☐ Combination of both ☐ Other (Please specify): *Click or tap here to enter text.*
33. How will you ensure that the keywords accurately reflect the content of the data?
Click or tap here to type your answer.
34. Will the metadata be made available through a metadata repository or catalogue that can be searched and indexed by search engines and other discovery services?
☐ Yes ☐ No
35. Will the metadata be freely available or restricted to certain users or communities?
☐ Freely available ☐ Restricted to certain users. ☐ Restricted to certain communities. ☐ Other (Please specify): *Click or tap here to type your answer.*

4.2 FAIR DATA: Making Data Accessible

4.2.1 Repository

36. Have you identified one or more trusted repositories to deposit the data and ensure its long-term preservation and accessibility? If yes please name them below.
Click or tap here to type your answer.
37. Have you confirmed that the identified repository meets the relevant standards and requirements for data management and preservation?
☐ Yes ☐ No
38. Will the repository assign a persistent identifier to the data, such as a DOI or URN, and ensure that the identifier resolves to the digital object?
☐ Yes ☐No

4.2.2 Data

39. Will all data generated by the project be made openly available, or will certain datasets be subject to restrictions?

☐ Openly available ☐ Certain datasets will be subject to restrictions. ☐ Other (Please specify): *Click or tap here to type your answer.*
40. If there are any restrictions on data access, what are the reasons for them, and how will they be enforced? (Select all that apply)
☐ Legal or ethical reasons ☐ contractual reasons ☐ There are no restrictions on data access. ☐ Other (Please specify): *Click or tap here to type your answer.*
41. how will these restrictions be enforced?
Click or tap here to type your answer.
42. Is an embargo period placed on any of the data? If so, what is the duration of the embargo and why is it needed?
Click or tap here to type your answer.
43. Will the data be accessible through a standardized access protocol, such as an API or web services?
☐ Yes ☐No
44. If access to the data is restricted, how will access be granted to approved users, both during and after the end of the project? (Select all that apply)
☐ Password-protected access ☐ User authentication process ☐ Access to data is not restricted. ☐ Other (Please specify): *Click or tap here to type your answer.*
45. What measures will be put in place to verify the identity of users accessing the data?
☐ User registration process ☐ Two-factor authentication ☐ Other (Please specify): *Click or tap here to type your answer.*
46. How will you ensure that your data remains accessible and usable over time, including any necessary updates to metadata or other documentation?
☐ Regularly update metadata or documentation ☐ Use version control. ☐ Other (Please specify): *Click or tap here to type your answer.*

4.2.3 Metadata

47. Will metadata describing the data be made openly available and licensed under a public domain dedication, such as CC0?
☐ Yes, metadata will be openly available and licensed under a public domain dedication, such as CC0. ☐ Metadata will be available, but not licensed under a public domain dedication such as CC0. ☐ No, metadata will not be openly available. ☐ Other (Please specify): *Click or tap here to type your answer.*
48. Will the metadata be structured in accordance with relevant disciplinary or general standards to facilitate discovery and interoperability?

☐ Yes, disciplinary standards will be followed. ☐ Yes, general standards will be followed. ☐ Both disciplinary and general standards will be followed. ☐ No, standards will not be followed. ☐ Other (Please specify): *Click or tap here to type your answer.*
49. Will the metadata include sufficient information to enable users to access the data, such as links to the data or instructions for accessing it?
☐ Yes, links to the data will be provided. ☐ Yes, instructions for accessing the data will be provided. ☐ Yes, both links and instructions will be provided. ☐ No, neither links nor instructions will be provided. ☐ Other (Please specify): *Click or tap here to type your answer.*
50. How long will the metadata and data be available and findable, and how will they be maintained over time?
Click or tap here to type your answer.
51. If software is required to read or access the data, how will you ensure it?
☐ Documentation to the software will be included. ☐ References to the software will be included. ☐ Both documentation and references to the software will be included. ☐ Not applicable, software is not required to access or read the data. ☐ Other (Please specify): *Click or tap here to type your answer.*
52. Will the software be made available as open-source code?
☐ Yes, the software will be made available as open-source code. ☐ No, the software will not be made available as open-source code. ☐ Not applicable, software is not required to access or read the data

4.2.4 Data Access Committee

53. Is there a need for a data access committee to evaluate and approve access requests to personal or sensitive data?
☐ Yes ☐ No
54. If there is a need for a Data Access Committee, what are the criteria for access and who will serve on the committee?
Click or tap here to type your answer.
55. How will the committee ensure that data access is granted in accordance with relevant legal and ethical requirements? (Select all that apply)
☐ User authentication process ☐ Data use agreement or license ☐ Conducting background checks on users requesting access ☐ Monitoring or auditing data use to ensure compliance with legal and ethical requirements. ☐ There is no need for a Data Access Committee ☐ Other (Please specify): *Click or tap here to type your answer.*

4.3 FAIR DATA - Making data interoperable.

56. Have you identified relevant data and metadata standards and best practices for ensuring data interoperability within and across disciplines?
☐ Yes ☐ No
57. Will you follow community-endorsed best practices for data interoperability, such as the FAIR principles?
☐ Yes ☐ No ☐ Other (Please specify): *Click or tap here to type your answer.*
58. What data and metadata vocabularies, standards, formats, or methodologies will you follow to make your data interoperable to allow data exchange and re-use within and across disciplines? (Select all that apply)
☐ Dublin Core ☐ EML (Ecological Metadata Language) ☐ FGDC (Federal Geographic Data Committee) metadata standard ☐ ISO/TC 211 metadata standard ☐ Other (Please specify): *Click or tap here to type your answer.*
59. If you need to use project-specific or uncommon ontologies or vocabularies, how will you ensure that they are interoperable with more commonly used ones? (Select all that apply)
☐ Provide mappings to more commonly used ontologies. ☐ Work with experts in the field to ensure compatibility. ☐ Not applicable ☐ Other (Please specify): *Click or tap here to type your answer.*
60. Will you provide mappings or translations between your ontologies and more commonly used ones to facilitate interoperability?
☐ Yes ☐ No
61. Will you openly publish the ontologies or vocabularies you generate to enable others to reuse, refine, or extend them?
☐ Yes ☐ No
62. Will your data include qualified references to other data (e.g. other data from your project, or datasets from previous research) to enable users to contextualize, understand, and re-use your data?
☐ Yes, all relevant data will be referenced and properly cited. ☐ Yes, but only the most important or relevant data will be referenced. ☐ No, the data will not include any references to other data
63. How will you ensure that the references are accurately recorded and that the linked data remain accessible over time? (Select all that apply)
☐ By using persistent identifiers such as DOIs or URNs ☐ By following relevant standards and best practices for data citation ☐ By providing detailed documentation on how to access the referenced data ☐ By ensuring the referenced data is deposited in trusted repositories with proper preservation and access policies ☐ Other (Please specify): *Click or tap here to type your answer.*

4.4 FAIR DATA - Increase Data re-use

64. How will you provide documentation needed to validate data analysis and facilitate data re-use? (Select all that apply)
☐ Through readme files ☐ Through codebooks ☐ Through data dictionaries ☐ Other (Please specify): *Click or tap here to type your answer.*
65. Will the documentation include information on data cleaning, methodology, and variable definitions to help others understand and reproduce your analysis?
☐ Yes ☐ No
66. How will you ensure that the documentation is clear, concise, and easily understandable by others? (Select all that apply)
☐ By hiring a technical writer ☐ By conducting usability testing ☐ By having non-experts review the documentation ☐ Other (Please specify): *Click or tap here to type your answer.*
67. Will you make your data freely available for maximum reuse and impact, and ensure compliance with relevant grant agreement obligations and standard reuse licenses such as Creative Commons?
☐ Yes, the data will be made freely available and licensed under Creative Commons or similar standard licenses. ☐ Yes, the data will be made freely available, but no license will be used. ☐ No, the data will not be made freely available. ☐ Other (Please specify): *Click or tap here to type your answer.*
68. Will the data produced in the project be useable by third parties, in particular after the end of the project?
☐ Yes ☐ No
69. How will you ensure that the data remains accessible and useable by third parties, including after the end of the project? (Select all that apply)
☐ By depositing the data in a trusted repository ☐ By providing permanent identifiers to the data ☐ By documenting the data and metadata according to standards ☐ Other (Please specify): *Click or tap here to type your answer.*
70. Will you provide sufficient documentation and metadata to enable others to understand and reuse the data?
☐ Yes ☐ No
71. How will you document the provenance of the data, including information on how it was collected, processed, and analysed?
☐ By using appropriate metadata standards and vocabularies ☐ By providing detailed documentation and version control information ☐ Other (Please specify): *Click or tap here to type your answer.*
72. What processes and measures will you use to ensure the quality, accuracy, completeness, consistency, and freedom from errors or bias of the data?

☐ Validation checks or error detection routines. ☐ Manual data review and verification ☐ Peer review or quality control procedures ☐ Statistical methods or software tools for data analysis and validation ☐ Other (Please specify): *Click or tap here to type your answer.*
73. Will you provide information on any limitations or assumptions associated with the data to help third parties interpret and use it appropriately?
☐ Yes ☐ No

5 Other research outputs

74. Which research outputs other than data will be generated and/or re-used throughout the project? (Select all that apply)
☐ Software ☐ Workflows ☐ Protocols ☐ Models ☐ Physical materials ☐ Other (Please specify): *Click or tap here to type your answer.*
75. How will you manage and preserve these outputs to ensure their long-term accessibility and usability? (Select all that apply)
☐ Store them in a trusted repository. ☐ Use appropriate metadata standards and documentation to describe and facilitate re-use. ☐ Use appropriate license to ensure that they can be freely reused and repurposed by others. ☐ Other (Please specify): *Click or tap here to type your answer.*
76. How will you ensure that research outputs other than data are managed and shared in line with the FAIR principles? (Select all that apply)
☐ Use appropriate metadata standards and documentation to describe and facilitate re-use ☐ Publish them using an appropriate license to ensure that they can be freely reused and repurposed by others. ☐ Make them available through a trusted repository or other public platforms. ☐ Other (Please specify): *Click or tap here to type your answer.*
77. How will you manage and share physical materials or other non-digital outputs?
Click or tap here to type your answer.
78. How will you allocate resources to support the management and sharing of research outputs other than data? ***What resources, such as storage, curation, or dissemination, will be necessary to support the management and sharing of these outputs? How will you ensure that sufficient resources are allocated to support the long-term preservation and accessibility of these outputs?***
Click or tap here to type your answer.

6 Allocation of resources

79. Have you identified the specific costs associated with making your data and other research outputs FAIR, including costs related to storage, curation, and dissemination?
Click or tap here to type your answer.
80. How will you allocate resources to support data management and sharing activities, such as storage, curation, and dissemination?
☐ By hiring dedicated data management personnel ☐ By allocating funds to purchase necessary hardware and software ☐ By using cloud-based storage and curation services ☐ Other (Please specify): *Click or tap here to type your answer.*
81. How will these costs be covered, and what funding sources will be used to support data management activities? *Will these costs be covered by the HORIZON EUROPE grant or other funding sources, and what are the conditions and limitations of these funding sources? How will you ensure that sufficient resources are allocated to support the necessary data management activities throughout the project lifecycle?*
Click or tap here to type your answer.
82. Who will be responsible for data management in your project, and what are their specific roles and responsibilities? *Have you identified a specific individual or team who will be responsible for data management activities, such as data curation, documentation, and dissemination? What are their specific roles and responsibilities, and how will they work with other project staff to ensure that data management activities are integrated into project workflows?*
Click or tap here to type your answer.
83. How will long-term preservation of data be ensured, and what resources will be necessary to accomplish this, including costs and potential value, who decides and how, what data will be kept, and for how long? *Have you identified a specific plan for the long-term preservation and accessibility of your data, including the necessary resources and infrastructure? Who will be responsible for deciding which data to keep and for how long, and what factors will be considered in these decisions? How will you ensure that the data is preserved and made accessible in a sustainable and responsible way, taking into account ethical and legal considerations?*
Click or tap here to type your answer.

7 Data Security

84. Have you identified the potential risks and threats to data security, such as unauthorized access or data loss, and developed a plan to mitigate these risks?
☐ Yes, potential risks and threats to data security have been identified, and a plan to mitigate these risks has been developed. ☐ No, potential risks and threats to data security have not been identified yet. ☐ Other (Please specify): *Click or tap here to type your answer.*
85. What are these potential risks and threats identified? (Select all that apply)
☐ Unauthorized access by hackers or cybercriminals ☐ Accidental data loss or corruption ☐ Data breaches resulting from inadequate security measures. ☐ Insider threats, such as employees mishandling data or deliberately leaking it ☐ Natural disasters or other environmental risks ☐ Cyber-attacks from foreign governments or state-sponsored actors ☐ Other (Please specify): *Click or tap here to type your answer.*
86. What is the plan to mitigate those risks?
Click or tap here to type your answer.
87. What provisions are or will be in place for data security, including measures for data recovery, secure storage and archiving, and transfer of data?
Click or tap here to type your answer.
88. Will the data be safely stored in trusted repositories for long-term preservation and curation?
☐ Yes ☐ No
89. Have you identified the relevant ethical and legal frameworks for data security and privacy, such as GDPR or other data protection regulations?
☐ GDPR (General Data Protection Regulation) ☐ HIPAA (Health Insurance Portability and Accountability Act) ☐ FERPA (Family Educational Rights and Privacy Act) ☐ Other (Please specify): *Click or tap here to type your answer.*
90. What measures will be in place to ensure compliance with these frameworks?
☐ Obtaining informed consent from participants ☐ Anonymizing or de-identifying sensitive data. ☐ Implementing access controls to limit data access to authorized personnel only. ☐ Other (Please specify): *Click or tap here to type your answer.*
91. How will you ensure that data security and privacy considerations are integrated into project workflows and activities throughout the project lifecycle?
☐ Including data security and privacy considerations in project planning and design ☐ Providing regular reminders or training on data security and privacy to staff ☐ Periodically reviewing and updating data security and privacy measures ☐ Other (Please specify): *Click or tap here to type your answer.*

92. Does your organization have appointed a Data Protection Officer
☐ Yes ☐ No ☐ Other (Please specify): *Click or tap here to type your answer.*
93. What measures will you put in place to ensure that data security and ethical considerations are addressed throughout the project lifecycle?
☐ By implementing appropriate access controls and encryption ☐ By conducting privacy impact assessments ☐ By complying with relevant ethical and legal requirements ☐ Other (Please specify): *Click or tap here to type your answer.*

8 Ethics

94. Have you identified any potential ethical or legal issues related to data sharing, such as privacy concerns or issues related to intellectual property rights? If so, how will these issues be addressed, and what measures will be in place to ensure compliance with relevant ethical and legal frameworks
Click or tap here to type your answer.

9 Personal Data

95. Will you collect/process any personal data? (If not, you can skip questions 96-110)
☐ Yes ☐ No
96. If yes, what personal data will you collect/process
Click or tap here to type your answer.
97. How will you collect the personal data?
Click or tap here to type your answer.
98. What is the purpose of the personal data procession and its relation to the objectives of the project? (Please specify, by showing that the data you will collect are proportional and limited to the objectives of the project following the data minimization principle)
Click or tap here to type your answer.
99. Will you process any "special categories of personal data" (racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic data, biometric data, health data, sex life or sexual orientation), and if so, for what purpose?
Click or tap here to type your answer.
100. Will informed consent be obtained from individuals providing personal data, and how will this consent be documented and stored?
☐ Yes, consent will be obtained and documented using a consent form. ☐ Yes, consent will be obtained and documented using other methods (please specify): *Click or tap here to type your answer.* ☐ No, consent will not be obtained. ☐ Not applicable to my research
101. Will informed consent be obtained from data subjects for data sharing and long-term preservation?
☐ Yes ☐ No ☐ Not applicable

102. If not, informed consent is requested and provided from the data subject, under which legal basis will you process personal data?
Click or tap here to type your answer.
103. Are you going to transfer /share personal data? If yes, where and how will you transfer them
Click or tap here to type your answer.
104. Are there any ethical or legal issues that can have an impact on data sharing?
Click or tap here to type your answer.
105. Will you safely store data in certified repositories for long time preservation and curation?
Click or tap here to type your answer.
106. Will you process any previously collected personal data (including use of pre-existing data sets or sources, merging existing data sets etc.) and under which legal basis.
Click or tap here to type your answer.
107. Have you identified the potential risks and threats to personal data security, such as unauthorized access or data loss, and developed a plan to mitigate these risks?
☐ Yes, potential risks and threats to personal data security have been identified, and a plan to mitigate these risks has been developed. ☐ No, potential risks and threats to personal data security have not been identified yet. ☐ Other (Please specify): *Click or tap here to type your answer.*
108. What provisions are or will be in place for data security, including measures for data recovery, secure storage and archiving, and transfer of personal data?
Click or tap here to type your answer.
109. Does your organization have a Personal Data Management Plan in place?
Click or tap here to type your answer.
110. Did you put in place any of the following measures to thoroughly implement the GDPR or equivalent legislation?
☐ Data protection Impact Assessment (DPIA) ☐ Designate a Data Protection Officer (DPO) ☐ Oversight mechanisms for data processing (limited access by qualifies members, mechanisms for logging data access and modifications) ☐ Measures to enhance privacy by design and default (e.g. anonymization, encryption, pseudonymization)

10 Artificial Intelligence

111. Will you be using A.I. tools or systems? ***If your answer is "No" please skip questions 112 - 116***
☐ Yes ☐ No
112. If yes, state what are these tools or describe the A.I. systems.
Click or tap here to type your answer.
113. Will the data holder be informed of the use of A.I. tools and systems prior?
☐ Yes ☐ No ☐ Not Applicable
114. Will there be continuous human supervision?
☐ Yes ☐ No ☐ Not Applicable
115. Have you identified the Risk Level in terms of the A.I. Act (Unacceptable, High, Limited, Minimal) for each A.I. tool and/or system that will be used? If Yes, please specify.
Click or tap here to type your answer.
116. Could the use of the A.I. tools or A.I. systems raise ethical concerns related to human rights and values and detail how this will be addressed?
Click or tap here to type your answer.

11 Other issues

117. Have you identified any relevant national, funder, sectorial, or departmental procedures for data management, and if so which ones? *Examples of national procedures for data management include: the General Data Protection Regulation (GDPR) in the European Union, the National Institutes of Health (NIH) Data Sharing Policy in the United States, and the Australian Code for the Responsible Conduct of Research.* *Examples of funder procedures for data management include: the National Science Foundation (NSF) Data Management Plan (DMP) requirements, the Wellcome Trust Data Management and Sharing policy, and the European Commission's Horizon 2020 Open Research Data Pilot.* *Examples of sectorial or departmental procedures for data management might include guidelines or policies from professional associations, research councils, or government agencies in a specific field or industry. For instance, the International Council for Science (ICSU) has developed data-sharing principles for scientific data, and the International Committee of Medical Journal Editors (ICMJE) requires authors to share data underlying published clinical trials.*
Click or tap here to type your answer.
118. What are the specific requirements or recommendations associated with each of these policies or procedures, and how will they be implemented in your project?
Click or tap here to type your answer.
119. How will you ensure that any external policies or procedures are integrated into project workflows and activities? Provide all applicable methods:
☐ Training ☐ Guidelines ☐ Other (Please specify): *Click or tap here to type your answer.*
120. Will you monitor and evaluate compliance with any external policies or procedures, and how will you ensure that any necessary updates or changes are made to project activities or workflows?
Click or tap here to type your answer.